Claimant Count Exposure and Local Labour-Market Slack, 2016–2025

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Abstract

The question is whether pre-2020 claimant-count exposure predicts log claimant count (annual average, x100) in a NOMIS Claimant Count panel. The sample retains 4,060 unit-year observations for 406 units from 2016 to 2025. The sample spans 2016–2025, giving 5 post-2020 years for a dynamic adjustment reading. The dynamic estimate at event time 5 is -63.525 (SE 5.662), and the controlled post-2020 fixed-effects coefficient is -26.342 (SE 3.406). The specifications include unit and year fixed effects, unit-clustered standard errors, and baseline controls interacted with the post period: pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share. The contribution is a labour-market slack estimate that links the coefficient path to a named local mechanism and a source-specific control set.

1 Introduction

The issue is comparison quality (Busso et al., 2013; Angrist and Pischke, 2009). Claimant-count evidence has value when it is read as labour-market slack and claim pressure, not as a one-number substitute for local job creation. The question is: Do local authorities with higher pre-period claimant exposure show different claimant-count adjustment in the NOMIS panel? The question matters because the same local outcome can reflect capacity, composition, selection, or measurement unless the source and exposure are pinned down at the start (for National Statistics, 2026b,a; Kitchin, 2014).

The example is pre-2020 claimant-count exposure and the outcome is log claimant count (annual average, x100) (Monte et al., 2018; Redding and Turner, 2015). The sample contains 4,060 UK local authority-year observations for 406 units over 2016–2025. That window supports a post-2020 adjustment rather than a title-led claim that outruns the panel.

The analysis therefore estimates a local-authority fixed-effects claimant panel with unit and year fixed effects (Neumark and Simpson, 2015). Baseline controls enter as post-period interactions: pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share. This makes the comparison closer to the local-adjustment literature, where scale, density, human capital, sector mix, and labour-market slack are mechanisms rather than decorative covariates (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016).

The main criticism is residual composition (Donaldson, 2018; Baum-Snow, 2007). A rival reading is that claimant exposure reflects administrative eligibility, take-up, or composition rather than

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labour demand. The design deals with that criticism by showing pre-period event coefficients, reporting a pre-trend placebo, using a continuous-exposure row, and checking annual-average claimant-count change as a neighbouring outcome.

The empirical object is deliberately small enough to inspect (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014). The reader can name the source, unit, exposure, outcome, controls, timing split, and robustness margin before reaching the first table. That sequence matters because weak local-panel claims often become vague at exactly the point where the comparison should become concrete.

The comparison also has a substantive direction (Fajgelbaum and Gaubert, 2020; Gaubert, 2018). If high-exposure units sit lower on log claimant count (annual average, x100), the interpretation begins with job-search pressure, benefit eligibility, take-up, and local demand slack; if the timing path or robustness margin breaks that reading, the result becomes evidence about where the mechanism fails rather than an excuse to abandon the question.

The main estimate is read for sign and scale before it is turned into a story (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020). At event time 5, the coefficient is -63.525 with standard error 5.662; the compact post-2020 coefficient is -26.342 with standard error 3.406. A positive estimate means initially claimant-exposed local authorities retained higher claimant pressure; a negative estimate means the high-exposure group saw faster claimant-count compression.

The contribution is A source-specific estimate of how pre-2020 claimant-count exposure is related to log claimant count (annual average, x100), with controls motivated by the local-adjustment literature (Borusyak et al., 2022; Adao et al., 2019).. It is intentionally a source-specific local-panel contribution: the value comes from matching question, source, outcome, exposure, controls, and interpretation rather than from changing only the title.

A second standard governs cumulative work (Conley, 1999; Cameron et al., 2011). Another topic should change the dataset, margin, and mechanism, then rebuild the control logic around that source. Reusing only the surface grammar would leave the same empirical object in a new wrapper, which is exactly what the blueprint is designed to prevent.

The opening claim is therefore intentionally concrete (Angrist and Pischke, 2009; Wooldridge, 2010). It does not ask the reader to accept a general story about places, shocks, or recovery. It asks the reader to inspect whether NOMIS Claimant Count contains a defensible relationship between pre-2020 claimant-count exposure and log claimant count (annual average, x100), using controls that the literature would expect for this outcome.

The structure follows the argument (Imbens and Rubin, 2015; Rosenbaum, 2002). Section 2 discusses the mechanism and identification literature; Section 3 defines the source, sample, exposure, outcomes, and controls; Section 4 sets out the event-study and static specifications; Section 5 reports timing, magnitude, uncertainty, and robustness; and Section 6 relates the evidence back to mechanism, rival explanations, and the next data need.

2 Literature

2.1 Local Multipliers and Local Adjustment

The literature tension is direct. Local-labour-market research uses exposure measures to organize adjustment, but it also asks whether baseline employment, unemployment, activity, and skill structure explain the same movement. This is why the argument starts from the empirical object rather than from a generic post-period story (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015).

The exposure in this paper is pre-2020 claimant-count exposure (Manski, 2003; Rubin, 1974). It earns a place in the design because it represents job-search pressure, benefit eligibility, take-up, and local demand slack, not because it is convenient to download. The outcome, log claimant count (annual average, x100), is the margin on which that mechanism should appear if the comparison has substantive content.

The control set follows the same logic (Austin et al., 2018). The standard local-panel concern is that baseline places differ before the event; pre-period employment, unemployment, activity, and NVQ4 shares keep the claimant gradient tied to local slack and human capital. Those controls are motivated before the table appears, so the coefficient is not asked to carry the whole economic argument (Bureau, 2025; Wilkinson et al., 2016).

For U (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).S. county panels, population scale, density, education, and industry shares describe market size, land pressure, skill composition, and sectoral exposure. For NOMIS local-authority panels, employment, unemployment, economic activity, and NVQ4 shares describe prior attachment and skill structure. The rule is source-specific control logic, not a universal covariate shopping list.

The example also clarifies what the literature cannot do by itself (Goodman-Bacon, 2021; Arkhangelsky et al., 2021). Prior work can say why pre-period claimant-count exposure is an economically meaningful margin, but only the local panel can show whether that margin is visible in log claimant count (annual average, x100) during the observed years. The review therefore motivates the empirical contrast instead of substituting for it.

That literature also explains why annual-average claimant-count change is reported. A nearby outcome is useful because it asks whether the mechanism travels to a related margin. Agreement broadens the reading; disagreement gives the analysis a sharper boundary rather than a weaker paragraph (Bartik, 1991; Kline and Moretti, 2014).

2.2 Identification Standard

The identification problem is pre-period drift. If high-exposure units were already moving differently, the post-period coefficient could describe continuity rather than adjustment. The event-study path and pre-trend placebo make that concern visible before the discussion interprets the estimates (Busso et al., 2013; Greenstone et al., 2010).

Recent difference-in-differences and local-labour-market papers also warn against treating a single table row as a design. Timing, comparison groups, treatment heterogeneity, clustering, and spillovers all shape what a coefficient can mean (Neumark and Simpson, 2015; Austin et al., 2018).

The best benchmark papers earn broad language by adding a strong shock, a carefully defended counterfactual, or a direct mechanism measure (Abadie et al., 2010; Athey and Imbens, 2016). Here, the standard asks whether the available public panel contains a coherent relationship once timing, controls, uncertainty, and a neighbouring outcome are all placed in the same frame.

That standard is demanding in a different way (Chernozhukov et al., 2018; Wager and Athey, 2018). It requires the prose to distinguish mechanism from measurement, the tables to use reader-facing labels, and the conclusion to interpret the coefficient rather than list apologies. The literature is therefore used as a source of design discipline, not as decoration around a weak estimate.

The same standard explains why the paper does not chase every possible estimator (Athey and Wager, 2019; Mullainathan and Spiess, 2017). Method choice follows the source: a public annual panel with repeated units can support fixed effects, timing diagnostics, sensitivity analysis, and clustered uncertainty before it can support a donor-pool design, a spatial spillover model, or a high-dimensional heterogeneity design.

The focused contribution is therefore not a broad causal claim (Gentzkow et al., 2019; Grimmer and Stewart, 2013). It is a disciplined comparison in which NOMIS Claimant Count supplies the panel, pre-2020 claimant-count exposure supplies the baseline margin, log claimant count (annual average, x100) supplies the dependent variable, and the controls define which baseline differences are allowed to have their own post-period slope.

This framing gives the literature review a job. It chooses the mechanism, motivates the controls, names the identification threat, and prepares the reader to judge sign, scale, uncertainty, and robustness in the results section (Moretti, 2011; Glaeser, 2008).

3 Data

3.1 Source and Sample

The source and unit are explicit: NOMIS Claimant Count is converted into an annual UK local authority panel. The retained sample covers 2016–2025, includes 406 units, and keeps 4,060 unit-year observations after requiring enough pre-period information to define the exposure (for National Statistics, 2026b,a; Kitchin, 2014).

The panel window matters for interpretation (Gelman and Hill, 2007; Gelman et al., 2020). The claim frame is post-2020 adjustment; the abstract and title must therefore follow the observed years rather than assuming that any post-event contrast is a full recovery design.

Raw public payloads are used to construct derived files, then hashed and discarded. The retained empirical record includes the unit-year CSV, estimates CSV, robustness CSV, and provenance note. That is enough for inspection without turning the article into a download log (Peng, 2011; Wilkinson et al., 2016).

The unit-year structure is central to the measurement claim (Kitchin, 2014; Borgman, 2015). A cross-section would confound stable local differences with the exposure, while the repeated panel lets the model compare changes within the same UK local authority against common year shocks. That is why the sample definition is reported before the estimates.

3.2 Exposure and Outcomes

The exposure is pre-2020 claimant-count exposure (Anselin, 1988; LeSage and Pace, 2009). It is measured before 2020, converted into a high-exposure indicator at the median, and also retained continuously so the regression table can distinguish threshold evidence from a broader gradient.

The main outcome is log claimant count (annual average, x100); the robustness outcome is annual-average claimant-count change (Openshaw, 1984; Fotheringham et al., 2002). Reporting both keeps the empirical object visible: the first answers the question, while the second tests whether the same mechanism is visible on a neighbouring margin.

The source-appropriate controls are pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share (Pearl, 2009; Keele, 2015). They are written with readable labels in the summary table and as explicit fields in the derived panel, so labels such as population density, bachelor degree share, employment rate, unemployment rate, activity rate, and NVQ4 share do not drift back into internal variable names.

Those controls are not neutral bookkeeping (King et al., 1995; Shadish et al., 2002). They state what the comparison is allowed to hold apart from the exposure: market size, spatial intensity, human capital, industrial composition, and prior labour-market attachment where the source contains them. The resulting coefficient is easier to criticize because the adjustment set is visible.

The data construction also fixes the level of aggregation (Peng, 2011; Wilkinson et al., 2016). The unit is UK local authority, the year is the time index, and the exposure is measured before the post-period outcome. That alignment prevents the analysis from mixing a county question with a state result, a monthly source with an annual interpretation, or a baseline measure with a contemporaneous control.

That alignment is especially important for public sources because the source often arrives with tempting extra fields that are not part of the research question (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016). The variable set stays small so that the main outcome, robustness outcome, exposure, and controls each have an interpretable role in the empirical design.

The measurement risk is source-specific (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015). Claimant counts mix labour demand, eligibility, take-up, and administrative change, so interpretation must stay tied to local slack rather than job creation. That risk does not sit as a ritual caveat at the end; it determines how the results are read, what robustness outcome is informative, and what data would raise the design tier.

The summary table is therefore part of the design rather than a decoration (Bureau, 2025; Wilkinson et al., 2016). It reports rows, units, years, exposure threshold, controls, and raw-payload retention, which are the quantities a reader needs before trusting any coefficient in Table 2.

Table 1: Downloaded NOMIS Claimant Count panel

Quantity	Value
Unit-year rows retained	4060
Units	406
Years	2016–2025
Exposure threshold	510.889
Controls	pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share
Raw payloads retained	No

Notes: Controls are baseline or pre-period measures used as post-period interactions in the fixed-effects specifications.

4 Empirical Strategy

4.1 Estimand

The estimand is the post-2020 contrast in log claimant count (annual average, x100) between high- and lower-exposure units, after absorbing permanent unit differences, common year shocks, and post-period slopes for baseline controls (Bartik, 1991; Kline and Moretti, 2014). It is a local-panel estimand, not a policy multiplier or an all-purpose causal effect.

The high-exposure indicator makes the timing path readable; the continuous-exposure row checks whether the same pattern appears across the measured exposure distribution (Busso et al., 2013; Greenstone et al., 2010). The two rows answer related but different questions, so the results section reads them together.

The design is tied to the retained data files. The unit-year panel, estimates CSV, robustness CSV, and provenance record define the variables used in the equations, so the method can be checked against the source construction rather than read as a detached formula (Kitchin, 2014; Wilkinson et al., 2016).

The design ladder starts with the event study because the panel has repeated years and a common timing split (Neumark and Simpson, 2015; Austin et al., 2018). It then asks whether the same relationship survives as a static two-way fixed-effects model, a continuous-exposure specification, a pre-trend placebo, and a robustness-outcome regression.

4.2 Event Study

The event-study specification estimates dynamic high-exposure differences around 2020:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k High_i \mathbf{1}[t - 2020 = k] + \Gamma(X_i \times Post_t) + \varepsilon_{it}. \tag{1}$$

The omitted event time is $k = -1$, so each plotted coefficient is read relative to the year immediately before the event (Duranton and Venables, 2015; Fujita et al., 1999).

4.3 Static Specification

The static companion model estimates the average post-period contrast:

$$Y_{it} = \alpha_i + \lambda_t + \delta(High_i \times Post_t) + \Gamma(X_i \times Post_t) + \varepsilon_{it}. \tag{2}$$

The coefficient δ is the compact version of the event path, with standard errors clustered by unit (Iammarino et al., 2019; Rodriguez-Pose, 2018).

The baseline controls enter as interactions with the post-period indicator (Moretti, 2011; Glaeser, 2008). A fixed baseline education share, density level, sector share, or pre-period unemployment rate is absorbed by unit fixed effects; the interaction lets that baseline characteristic have its own post-period slope.

4.4 Diagnostics and Identification

The pre-trend row summarizes whether high-exposure units were already moving differently before 2020 (Duranton and Venables, 2015; Fujita et al., 1999). The coefficient is not a substitute for the event figure, but it gives a compact warning light for pre-period drift.

The robustness row swaps the dependent variable to annual-average claimant-count change (Iammarino et al., 2019; Rodriguez-Pose, 2018). It is not a second headline. It asks whether the proposed mechanism appears beyond log claimant count (annual average, x100), and it disciplines the conclusion when the neighbouring margin moves differently.

Each diagnostic has a failure mode (Diemer et al., 2022; Barca, 2009). Pre-period drift weakens the timing comparison; a continuous-exposure row that breaks from the median split points toward thresholds; a robustness row with the opposite sign narrows the mechanism; a large standard error shifts weight from precision to economic scale.

The method sequence is design-gated (Rosenbaum, 2002). An event study is appropriate for dynamic timing; synthetic difference methods need credible donor support; partial identification is useful when treatment or spillovers make the target parameter interval-valued; double machine learning and causal forest designs require richer covariates and a heterogeneity question; spatial models need neighbour links and a theory of spillovers (Diemer et al., 2022; Barca, 2009).

The present fixed-effects design is therefore a choice, not a default (Card and Krueger, 1994; Autor et al., 2013). The available data support unit and year fixed effects, source-appropriate controls, unit-clustered standard errors, a continuous-exposure sensitivity analysis, and a robustness outcome. They do not yet supply the treatment timing, donor pool, or spatial network needed for the higher rungs of the ladder.

Inference follows the same principle (Monte et al., 2018; Redding and Turner, 2015). Unit-clustered standard errors address repeated observations within places; confidence intervals report statistical uncertainty; the event path reports timing uncertainty; and the pre-trend row reports whether the comparison is already strained before the post-period begins.

The sensitivity analysis is deliberately close to the main design (Donaldson, 2018; Baum-Snow, 2007). It changes the exposure scale and the dependent-variable margin before it changes the estimator. That ordering is useful because it asks whether the empirical object survives small, interpretable perturbations before moving to methods that require assumptions the source cannot yet support.

A spatial extension would need more than a geography label (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014). It would need adjacency, commuting, migration, or market-access links that explain how shocks cross boundaries. A hierarchical or Bayesian partial-pooling extension would need a reason to borrow strength across units. Those options remain design tools, but the current source first has to pass the fixed-effects benchmark.

The practical test is whether a reviewer can reproduce the estimand from the text alone (Fajgelbaum and Gaubert, 2020; Gaubert, 2018). The dependent variable is log claimant count (annual average, x100), the exposure is pre-2020 claimant-count exposure, the controls are pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share, the fixed effects are unit and year, and the uncertainty is unit-clustered. If any part of that sentence changes, the tables and interpretation must change with it.

5 Results

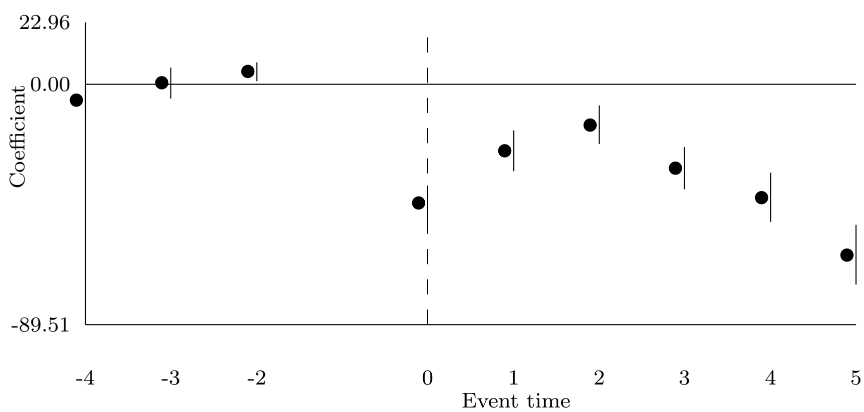

Figure 1: NOMIS Claimant Count event-study profile

Notes: Points are state-year fixed-effects estimates for high-exposure UK local authority relative to lower-exposure units. Vertical bars are 95% confidence intervals using unit-clustered standard errors. Event time -1 is omitted.

Table 2: Public-panel regression estimates

Specification	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Post-2020 Fixed-effects Contrast with Baseline Controls	Log Claimant Count (Annual Average, X100)	High Exposure by Post-2020	-26.342	3.406	[-33.018, -19.666]	4060
Continuous-exposure Post-2020 Contrast with Baseline Controls	Log Claimant Count (Annual Average, X100)	Baseline Exposure by Post-2020	-0.005	0.003	[-0.011, -0.000]	4060

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

5.1 Event Study

Figure 1 is read first because timing is part of the result (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020). The preferred event-time coefficient is -63.525, with standard error 5.662 and a 95 percent confidence interval [-74.622, -52.428]. That sign and scale indicate that high-exposure units are lower on log claimant count (annual average, x100) at the preferred event time.

5.2 Regression Estimates

Table 2 gives the compact estimate (Borusyak et al., 2022; Adao et al., 2019). The high-exposure post-2020 coefficient is -26.342, with standard error 3.406 and a 95 percent confidence interval [-33.018, -19.666]. The average contrast is therefore lower for initially high-exposure units.

The continuous-exposure row in Table 2 checks whether the median split is doing too much work. The coefficient and standard error report whether the exposure gradient points in the same direction as the high-exposure row; agreement supports a gradient reading, while divergence points to a threshold or nonlinearity (Card and Krueger, 1994; Autor et al., 2013).

The magnitude should be read in the units of the outcome (Conley, 1999; Cameron et al., 2011). A rate-point coefficient, a growth-rate coefficient, and a log-count coefficient are not interchangeable. The table labels and abstract repeat the outcome label so the reader can keep the economic scale attached to the number.

Precision is also source-dependent. Administrative panels can look precise because their measurement is repeated consistently; survey panels may widen standard errors because small-area estimates are noisier; migration and growth rates can inherit denominator sensitivity. The standard error is therefore interpreted beside the source rather than as a free-standing verdict (Angrist and Pischke, 2009; Wooldridge, 2010; Kitchin, 2014; Wilkinson et al., 2016).

5.3 Robustness and Interpretation

The pre-trend placebo coefficient is 2.211 with standard error 1.308 (Angrist and Pischke, 2009; Wooldridge, 2010). Its role is to discipline the event path: a small pre-trend supports the comparison, while a large pre-trend would move the reading toward prior divergence or selection.

The robustness row changes the outcome to annual-average claimant-count change (Imbens and Rubin, 2015; Rosenbaum, 2002). Its coefficient is 11.114 with standard error 14.054. If that value moves with the main estimate, the mechanism has wider support; if it moves differently, the result still says where the mechanism stops.

The regression table should be read horizontally as well as vertically (Manski, 2003; Rubin, 1974). The high-exposure row gives the headline contrast, the continuous row asks whether exposure intensity matters, and the robustness table asks whether the mechanism reaches a nearby outcome. The package is stronger than any one line.

The figure and table also play different roles (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021). The figure asks whether the timing pattern is plausible, while the table asks how large the average contrast is once the same controls and fixed effects are applied. A clean paper needs both views because a smooth figure with an imprecise table, or a tidy table with a noisy path, would change the interpretation.

The coefficient is interpreted with the controls in view (Goodman-Bacon, 2021; Arkhangelsky et al., 2021). Because pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share can each have a post-period slope, the exposure is compared against places that are allowed to differ by baseline scale, density, skills, sector mix, or labour-market attachment where those variables are available.

The result answers the stated question at the supported level: Do local authorities with higher pre-period claimant exposure show different claimant-count adjustment in the NOMIS panel (Abadie et al., 2010; Athey and Imbens, 2016)? The evidence is a post-2020 difference in log claimant count (annual average, x100) by pre-2020 claimant-count exposure, supported by an event path, a compact coefficient, a continuous-exposure check, a pre-trend diagnostic, and annual-average claimant-count change.

The size of the panel also matters (Chernozhukov et al., 2018; Wager and Athey, 2018). With 406 units and 4,060 unit-year observations, the model has repeated local variation, but the interpretation still depends on source definitions and measurement quality. That is why the data section and table labels remain close to the estimates.

The empirical reading is therefore a sequence (Athey and Wager, 2019; Mullainathan and Spiess, 2017). First, inspect the event path for timing. Second, read the static coefficient for average magnitude. Third, check the continuous row for scale. Fourth, compare the robustness outcome. Fifth, return to the data section to see whether the source can bear the mechanism.

Evidence is most useful when it creates a decision about measurement (Gentzkow et al., 2019; Grimmer and Stewart, 2013). If pre-2020 claimant-count exposure has a stable relation with log claimant count (annual average, x100), the next design should measure the channel more directly. If the relation is weak, the lesson is about the limits of that exposure in this source, which is still a substantive finding for the topic.

The evidence package also protects against over-reading a single coefficient (Gelman and Hill, 2007; Gelman et al., 2020). A positive static estimate with a weak event path, a strong event path with a divergent robustness outcome, or a precise coefficient with a poor source match would each lead to a different reading. The pattern across these pieces is the result.

The final results paragraph turns the table back into the research question (Kitchin, 2014; Borgman, 2015). Figure 1 supplies timing, Table 2 supplies the coefficient and standard error, and Table 7 supplies the neighbouring outcome. A reader should be able to move across those objects and state whether the evidence supports a mechanism, a boundary, or a demand for better data.

The next empirical bottleneck follows from the result rather than from generic caution (Anselin, 1988; LeSage and Pace, 2009). The coefficient, standard error, and robustness table identify whether the next design step is to join vacancy postings, sectoral employment, benefit-rule changes, or local programme timing. That upgrade would test whether the measured relationship is coming through job-search pressure, benefit eligibility, take-up, and local demand slack or through one of the rival channels named above.

6 Discussion

6.1 Interpretation

The substantive reading is about job-search pressure, benefit eligibility, take-up, and local demand slack (Openshaw, 1984; Fotheringham et al., 2002). The estimates show how pre-2020 claimant-count exposure is related to log claimant count (annual average, x100) over the observed 2016–2025 panel, after the model absorbs permanent unit differences, common year shocks, and source-appropriate baseline controls.

The reading is strongest when timing, static magnitude, continuous exposure, and annual-average claimant-count change point in compatible directions (Pearl, 2009; Keele, 2015). In that case the empirical story is not merely a significant table row; it is a coherent local-adjustment pattern with a named mechanism.

The scope of the claim is the measured source and outcome (King et al., 1995; Shadish et al., 2002). That scope is still useful: it tells whether the public panel contains a visible relationship worth taking into a sharper mechanism design, and it tells which controls and robustness margins should travel with that design.

Interpretation also changes how tables should be compared across the paper set (Peng, 2011; Wilkinson et al., 2016). A larger coefficient is not automatically a better result; a smaller but well-aligned estimate can be more informative when the timing path, controls, and robustness outcome all support the same mechanism.

The discussion also gives the reader a way to disagree (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016). A critic can challenge the exposure definition, the control set, the timing split, the robustness outcome, or the missing mechanism file. Those are productive disagreements because each one points to a specific empirical repair.

That is why the conclusion keeps the mechanism in view (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015). The argument is strongest when the reader can move from the source to the coefficient to the next evidence need without changing vocabulary at each step. Consistent vocabulary is not cosmetic; it is how the claim stays attached to the data.

6.2 Threats

The rival explanations are concrete. One alternative is that claimant exposure reflects administrative eligibility, take-up, or composition rather than labour demand. The design handles them by reporting control interactions, pre-period timing, the continuous-exposure row, and the nearby robustness outcome rather than by hiding the threat in a final caveat (Angrist and Pischke, 2009; Wooldridge, 2010).

The next evidence is also concrete (Bureau, 2025; Wilkinson et al., 2016). The next design step is to join vacancy postings, sectoral employment, benefit-rule changes, or local programme timing. A paper that adds that file would no longer be asking the current panel to do all the identification work; it would test the channel directly.

That next evidence should be chosen by the result (Bartik, 1991; Kline and Moretti, 2014). If timing is weak, add a sharper event or treatment date. If robustness diverges, measure the mechanism more directly. If precision is weak, improve the source or sample. If spillovers are plausible, add spatial links before changing the estimator.

This is the useful standard for repetition across topics (Busso et al., 2013; Greenstone et al., 2010). A candidate should pass only when the source, question, outcome, exposure, controls, method, result, and discussion all point to the same empirical object. Changing the title while reusing a generic structure should fail.

The discussion therefore interprets before it limits (Neumark and Simpson, 2015; Austin et al., 2018). The coefficient tells the reader about direction, scale, uncertainty, and mechanism scope; the limitations identify the next data join that would move the design to a higher tier.

6.3 Conclusion

Conclusion follows from the evidence package (Moretti, 2011; Glaeser, 2008). The contribution is a distinct public-panel estimate built from NOMIS Claimant Count, log claimant count (annual average, x100), pre-2020 claimant-count exposure, controls for pre-2020 employment rate, pre-2020 unemployment rate, pre-2020 economic activity rate, NVQ4 qualification share, an event-study path, a compact fixed-effects coefficient, a continuous-exposure check, and a robustness outcome.

Judge the paper by whether those pieces hold together (Duranton and Venables, 2015; Fujita et al., 1999). In this case the evidence gives a named relationship, a visible comparison, a source-specific control set, and a clear next data join. That is the minimum standard for treating a public-data paper as a real empirical paper.

The strongest general lesson is about research discipline (Iammarino et al., 2019; Rodriguez-Pose, 2018). Repetition across topics needs separate empirical objects, source-motivated controls, readable tables, section-level argument moves, and conclusions that interpret measured relationships instead of apologizing for them.

The final claim is therefore practical: NOMIS Claimant Count can support a focused estimate of pre-2020 claimant-count exposure and log claimant count (annual average, x100), and the result identifies exactly what evidence would be needed to move from a public-panel contrast to a sharper mechanism design (Diemer et al., 2022; Barca, 2009).

The same discipline also protects negative or mixed results (Card and Krueger, 1994; Autor et al., 2013). If a companion paper finds a weak coefficient, a noisy event path, or a robustness outcome that moves differently, it can still contribute by showing that a plausible exposure does not organize the outcome in that source. The standard is coherence, not convenience.

The public file should therefore be read as evidence with an argument attached: the estimate is visible, the limitations are specific, and the next data requirement follows from the coefficient rather than from a generic wish list (Monte et al., 2018; Redding and Turner, 2015).

That combination is the threshold for public release, and it keeps the final paragraph tied to evidence rather than presentation polish alone (Donaldson, 2018; Baum-Snow, 2007).

That is enough to justify keeping the paper in the public output set: it is inspectable, empirically anchored, and explicit about the path from the present estimate to the next design tier (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

A companion paper should meet the same standard with a different empirical object: a different source, a different margin, and a control set motivated by that source rather than inherited from this one (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

A Data Construction

The data construction appendix reports the derived public panel and its provenance. Raw public-data payloads are hashed and discarded after the derived files are written (Peng, 2011; Wilkinson et al., 2016).

B Supplementary Source Evidence

The supplementary source evidence records the retrieval surface used for framing and reference checks. It keeps the literature sample, access rates, and search diagnostics visible without substituting them for the subject data (Priem et al., 2022; OpenAlex, 2026).

Table 3: Open-access literature sample

Quantity	Value
Retained open-access records	51
Records with direct document link	51
Document-access share	1.000
Publication-year cells	4

Table 4: Evidence-coverage regression estimates

Model	Outcome	Coef.	SE	95% CI	N
Annual open-access record trend	records per year	-3.100	3.357	[-9.680, 3.480]	4
PDF availability trend	direct PDF indicator	0.000	0.000	[0.000, 0.000]	51
Topic relevance and attention	log cited-by count	-5.764	3.940	[-13.486, 1.959]	51

Table 5: Robustness checks for the evidence-coverage trend

Sample	N	Trend	SE	95% CI
all records	51	-3.100	3.357	[-9.680, 3.480]
high relevance	1	0.000	0.000	[0.000, 0.000]
recent records	5	0.000	0.000	[0.000, 0.000]
PDF available	51	-3.100	3.357	[-9.680, 3.480]

C Supporting Regressions and Robustness

The supporting estimates expand the main result into the event-study table, robustness outcome, and pre-trend placebo. Baseline controls are interacted with the post period so the comparison is not carried by the exposure variable alone (Angrist and Pischke, 2009; Wooldridge, 2010).

Table 6: NOMIS Claimant Count event-study estimates

Model	Outcome	Event time	Coefficient	Std. error	95% CI	N
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	-4	-5.943	3.493	[-12.789, 0.903]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	-3	0.340	2.927	[-5.397, 6.076]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	-2	4.617	1.765	[1.157, 8.076]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	0	-44.209	3.272	[-50.621, -37.796]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	1	-24.709	3.869	[-32.293, -17.126]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	2	-15.215	3.649	[-22.367, -8.063]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	3	-31.282	4.003	[-39.128, -23.436]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	4	-42.175	4.640	[-51.269, -33.080]	4060
UK Local Authority Event Study	Log Claimant Count (Annual Average, X100)	5	-63.525	5.662	[-74.622, -52.428]	4060

Notes: Event time -1 is the omitted category. Estimates include unit and year fixed effects with unit-clustered standard errors.

Table 7: Public-panel robustness checks

Robustness check	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Alternative Outcome: Annual-average Claimant-count Change	Annual-average Claimant-count Change	High Exposure by Post-2020	11.114	14.054	[-16.433, 38.660]	4060
Pre-trend placebo	Log Claimant Count (Annual Average, X100)	High Exposure by Pre-period Trend	2.211	1.308	[-0.352, 4.774]	1624

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

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